MATH102BKK · HARBOUR.SPACE BANGKOK

Homework 4 — Function families

Session 4 · Polynomials, exponentials, logarithms, and trigonometric models

Set
Thu 10 Sep 2026
Due
Fri 11 Sep 2026, start of class
Marks
24

HOW THIS IS MARKED Every answer carries one sentence of justification. A correct value with no reasoning scores half. A wrong value with a clear, nearly-correct argument scores most of the marks. You may discuss problems with anyone and may use AI tools — but you must be able to explain every line you submit.

BEFORE YOU START Conditions are part of an answer. Before using a logarithm, state which inputs are allowed. When you compare growth, say *for sufficiently large* n; one numerical example does not prove a growth claim.

Part A · Mechanics (8 marks)

1 POLYNOMIAL SHAPE — 2 marks

Let $p(x) = -3x^4 + 2x - 7$.

a. State the degree and leading coefficient of p.

b. Describe what the graph does at its left and right ends.

Give one sentence explaining why the lower-power terms do not change your answer to part (b).

SOLUTION The degree is 4 and the leading coefficient is -3. The degree is even and the leading coefficient is negative, so the graph goes down at both ends: $p(x) \to -\infty$ as $x \to -\infty$ and as $x \to \infty$. When $|x|$ is large, the leading term $-3x^4$ is much larger in size than $2x - 7$, so it determines the end behaviour. 1 mark for degree + leading coefficient; 1 mark for both ends with a reason.

2 QUADRATIC TRANSFORMATION

2 marks

Start with $f(x) = x^2$. The function

$$g(x) = -(x-2)^2 + 4$$

is made from f.

a. Describe the shift and reflection.

b. Give the vertex and find the real roots of g.

> **SOLUTION** Move the graph of $y = x^2$ right by 2, reflect it in the x-axis, then move it up by 4. Its vertex is $(2, 4)$. For roots, $-(x-2)^2 + 4 = 0 \iff (x-2)^2 = 4 \iff x - 2 = \pm 2$, so $x = 0$ or $x = 4$. 1 mark for the three transformations; 1 mark for the vertex and both roots.

3 EXPONENTIAL EQUATION

2 marks

Solve $5^{2x-1} = 125$. Rewrite 125 with base 5, then state why you may set the exponents equal.

> **SOLUTION** Since $125 = 5^3$, we have $5^{2x-1} = 5^3$. The base 5 is positive and is not 1, so the exponential function 5^t is one-to-one and $2x - 1 = 3$. Hence $x = 2$. 1 mark for $x = 2$; 1 mark for matching the bases and stating the condition.

4 CHANGE OF BASE

2 marks

Use the change-of-base formula to calculate $\log_4 32$. Show the formula you use and give the exact answer.

> **SOLUTION**
>
> $$\log_4 32 = \frac{\ln 32}{\ln 4} = \frac{\ln(2^5)}{\ln(2^2)} = \frac{5 \ln 2}{2 \ln 2} = \frac{5}{2}.$$
>
> The formula is valid because the input 32 is positive and the bases 4 and e are positive and not equal to 1. 1 mark for $\frac{5}{2}$; 1 mark for a correct use of the formula and its conditions.

Part B · Reasoning and modelling (11 marks)

5 LOGARITHMIC EQUATION 3 marks

Solve over the real numbers:

$$\log_2(x-1)+\log_2(x+1)=3.$$

State the domain before combining the logarithms. Check every possible answer in the original equation.

SOLUTION Both logarithms require positive inputs: $x-1>0$ and $x+1>0$. Therefore $x>1$. On this domain, the product law gives

$$\log_2\big((x-1)(x+1)\big)=3 \iff x^2-1=8 \iff x^2=9.$$

The algebra gives $x=3$ or $x=-3$, but the domain keeps only $x=3$. Check: $\log_2(2)+\log_2(4)=1+2=3$. 1 mark for the domain; 1 mark for legal log manipulation and algebra; 1 mark for rejecting -3 and checking 3.

6 GROWTH COMPARISON 3 marks

Put the following functions in order from slowest growth to fastest growth as the positive integer n becomes very large:

$$\log_2 n, \qquad n, \qquad n\log_2 n, \qquad n^2, \qquad 2^n.$$

In two or three sentences, explain why checking one value such as $n=3$ does not prove the full ranking.

SOLUTION The eventual order is

$$\log_2 n<n<n\log_2 n<n^2<2^n.$$

This is an eventual statement: it becomes true once n is large enough, not necessarily at every positive integer. For example, at $n=3$, $2^n=8<9=n^2$, but for sufficiently large n the exponential 2^n is larger than n^2. 1 mark for the complete ranking; 1 mark for saying "eventually" or its meaning; 1 mark for a valid explanation using $n=3$.

7 PERIODIC MODEL 5 marks

Let t be the number of hours after midnight. A simple model for temperature is

$$T(t) = 3\sin\left(\frac{\pi}{6}(t-1)\right) + 20 \qquad (0 \le t \le 12).$$

a. State the amplitude, midline, period, and range.

b. At what time does the model first reach a maximum? At what time does it reach a minimum?

c. Explain why $(t-1)$ moves the graph right by 1 hour, rather than left by 1 hour.

> **SOLUTION** The amplitude is 3, the midline is $y = 20$, the period is $\dfrac{2\pi}{\pi/6} = 12$ hours, and the range is $[17, 23]$. A maximum occurs when the sine input is $\pi/2$, so $\frac{\pi}{6}(t-1) = \frac{\pi}{2}$ gives $t = 4$. A minimum occurs when the input is $3\pi/2$, giving $t = 10$. In $f(t-1)$, the new input must be $t = 1$ to give the old value $f(0)$, so the graph moves right by 1. 2 marks for amplitude, midline, period, and range; 2 marks for both times with reasoning; 1 mark for the shift explanation.

Part C · Communicate precisely (5 marks)

8 FIND THE ERROR 3 marks

A student writes: "$\log_2(u+v) = \log_2 u + \log_2 v$ because logarithms turn an operation into addition."

a. Identify the false statement.

b. Give a counterexample with positive u and v.

c. Write the correct logarithm law for a product, including its condition.

> **SOLUTION** The false statement is $\log_2(u+v) = \log_2 u + \log_2 v$. Take $u = v = 1$: $\log_2(1+1) = \log_2 2 = 1$, while $\log_2 1 + \log_2 1 = 0 + 0 = 0$. The correct product law is $\log_2(uv) = \log_2 u + \log_2 v$ for $u > 0$ and $v > 0$. 1 mark for identifying the false law; 1 mark for the complete counterexample; 1 mark for the product law with $u, v > 0$.

9 MAKE A CLAIM PRECISE 2 marks

The statement "An exponential function is always bigger than a polynomial" is false or unclear as written. Rewrite it as a true statement about c^n and n^k, including all important conditions. Then give one numerical example showing why the words "for sufficiently large n " are necessary.

SOLUTION One correct version is: "For every fixed $c > 1$ and $k > 0$, there is a positive integer N such that $c^n > n^k$ for every integer $n \geq N$." The word "fixed" matters: the bases and powers are not changing with n. For $c = 2$, $k = 2$, and $n = 3$, we have $2^3 = 8 < 9 = 3^2$, so the inequality is not true for every positive n. 1 mark for a correct eventual-growth statement with $c > 1$, $k > 0$, and fixed quantities; 1 mark for a valid counterexample to "always".